MIT 18.06

Lecture 1 : The Geometry of linear Equations

→ The Goal of this lecture is to understand the system $Ax = b$ from two different perspectives.

① Row Picture

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Each equation in the system represents a line (in 2D) or a plane (in 3D). The solution is the point where these lines or planes intersect

② Column Picture (Crucial for AI)

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Instead of looking at rows, we look at the columns of matrix A. We view the system as a linear combination of vectors.

Let's take an example of 2D first!

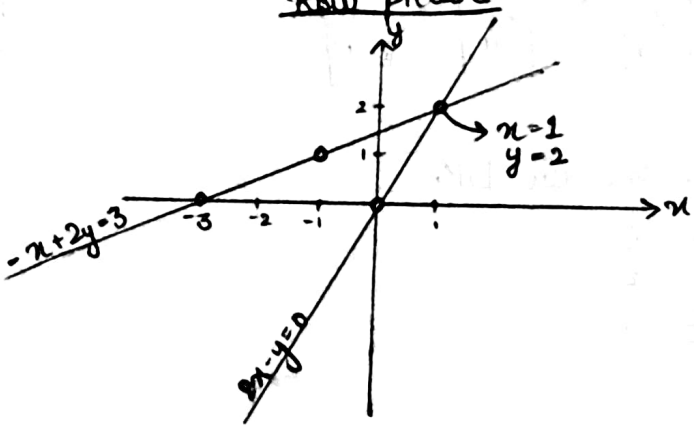
$$2x - y = 0$$
$$-x + 2y = 3$$

Matrix form :

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A \quad x = b$$

Row picture



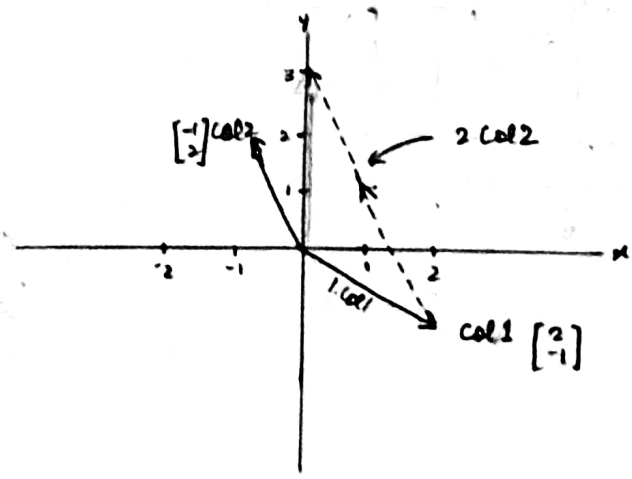
Linear combination of columns :

$$x\begin{bmatrix} 2 \\ -1 \end{bmatrix} + y\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Combination we take :

$x = 1$
$y = 2$

Column picture



All possible outcomes :

Type	Meaning	Geometry
Unique solution	Independent System	Intersect at a point
No solution	Inconsistent	Parallel lines
Infinite solution	Dependent	Same line

⊛ In a linear combination of columns we are looking for numbers x, y that scale the column vector so they add up to the result vector b.

Now let's take an example of 3D!

$$2x - y = 0$$
$$-x + 2y - z = -1$$
$$-3y + 4z = 4$$

Matrix form:

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

$$A \quad x = b$$

Linear Combination of columns:

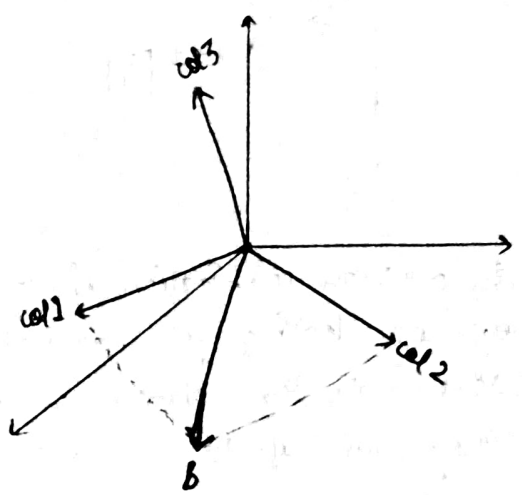
$$x\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z\begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

Combination we take:

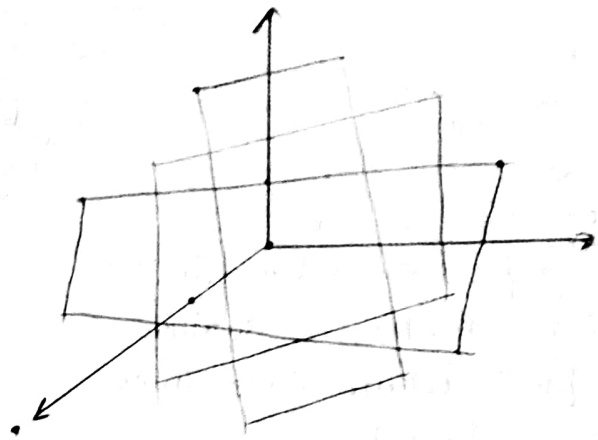
$x = 0$
$y = 0$
$z = 1$

If we take different b:

$$\underbrace{x\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z\begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}}_{\text{Column picture}}$$

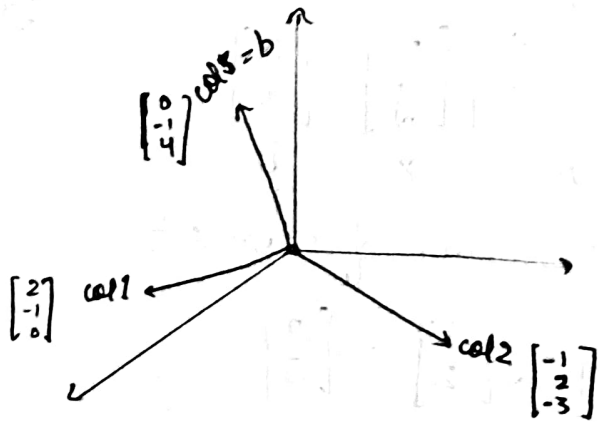


Row picture:



In row picture, we get value in different planes, so it is difficult to find the point where these planes meet.

Column Picture:



Q Can we solve $Ax = b$ for every b?

→ If all combination lie in one plane, the matrix cannot cover the whole space. So $Ax = b$ works only when b is in that plane, not for every b, as these are singular matrix.

Q Do the linear combination of column fill 3D space?

→ For matrix A, the answer is YES. These are non singular matrix.

Summary:
- Linear algebra is about structure not just solving
- The matrix form reveals relationships you can't easily see in equations